LOCAL CKN CONCENTRATION AND THE TYPE I/TYPE II DICHOTOMY FOR NAVIER–STOKES

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Abstract. We prove the local analytic reduction that precedes any classification of a three-dimensional Navier–Stokes singularity by blowup rate. For a suitable weak solution, a singular point at time T must carry a positive amount of the pressure-normalized Caffarelli–Kohn–Nirenberg critical density on arbitrarily small backward parabolic cylinders. If that density vanishes at each point of the singular set, the solution is regular at time T by the epsilon-regularity theorem. After positive local concentration has been obtained, the sequence either satisfies a Type I scale-invariant bound or it constitutes the local Type II alternative.

1. Introduction

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u + \nabla p = \Delta u, \qquad \nabla \cdot u = 0.$$

We use the standard normalization in which the viscosity is one. We work with suitable weak solutions in the sense of Caffarelli, Kohn, and Nirenberg [1], in the setting of Leray's weak solution theory [2]. This paper proves the local reduction from a possible singularity at time T to the Type I/Type II blowup-rate dichotomy.

The argument has four parts. First, the pressure-normalized critical quantities C and D are finite for suitable weak solutions and invariant under the usual parabolic scaling. Second, the Caffarelli–Kohn–Nirenberg epsilon-regularity theorem implies that vanishing of $C + D$ at a candidate singular point gives local regularity. Third, a finite-time singularity therefore produces a sequence of shrinking cylinders on which $C + D$ is bounded below by a positive constant. Finally, once such a sequence is fixed, it either satisfies a Type I scale-invariant bound or falls in the local Type II alternative.

The reduction is purely local. It does not use profile decompositions, global compactness theorems, decay statements, or Liouville theorems.

2. Suitable Weak Solutions and Singular Points

Definition 2.1 (Suitable weak solution)**.** Let $I \subset \mathbb{R}$ be an open interval. A pair (u, p) is a suitable weak solution on $\mathbb{R}^3 \times I$ if

$$u \in L^\infty_t L^2_{x,\mathrm{loc}} \cap L^2_t H^1_{x,\mathrm{loc}}, \qquad p \in L^{3/2}_{\mathrm{loc}}, \qquad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold in the sense of distributions, and the local energy inequality holds: for almost every $t_1 < t_2$ in I and every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times I)$,

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|u(x,t_2)|^2}{2}\phi(x,t_2)\,dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} |\nabla u|^2 \phi\,dx\,dt \\
&\le \int_{\mathbb{R}^3} \frac{|u(x,t_1)|^2}{2}\phi(x,t_1)\,dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} \frac{|u|^2}{2}(\partial_t\phi + \Delta\phi)\,dx\,dt \\
&\quad + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p\right) u\cdot\nabla\phi\,dx\,dt.
\end{aligned}$$

The pressure is understood modulo functions of time.

For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

We also write $Q_r(x_0, t_0)$ for $Q_r((x_0, t_0))$.

Definition 2.2 (Singular set at a fixed time)**.** For a time T, define

$$\Sigma(T) = \{x \in \mathbb{R}^3 :\ u \text{ is not locally bounded in any } Q_r(x,T)\}.$$

Equivalently, $x \notin \Sigma(T)$ if there are $r > 0$ and $M < \infty$ such that $\|u\|_{L^\infty(Q_r(x,T))} \le M$.

If a finite-time singularity occurs at time T, then $\Sigma(T) \neq \emptyset$ in this local sense.

3. Parabolic Rescaling and Critical Quantities

Fix $z_0 = (x_0, T)$. For $r > 0$, define the parabolic rescaling

$$u^{(r)}(y,s) = r\,u(x_0 + ry, T + r^2 s), \qquad p^{(r)}(y,s) = r^2 p(x_0 + ry, T + r^2 s).$$

The rescaled variables are defined on the set of (y, s) for which $(x_0 + ry, T + r^2 s)$ belongs to the original spacetime domain.

Proposition 3.1 (Invariance under parabolic rescaling)**.** *If (u, p) is a suitable weak solution near (x_0, T), then $(u^{(r)}, p^{(r)})$ is a suitable weak solution on the rescaled domain. The local energy inequality is preserved, and the pressure remains defined modulo functions of the rescaled time.*

Proof. The distributional equations and the divergence condition follow from the change of variables $x = x_0 + ry$, $t = T + r^2 s$. The local energy inequality is obtained by testing the original inequality with the corresponding rescaled nonnegative test function. The pressure transformation is the one dictated by Navier–Stokes scaling, and addition of a function of t becomes addition of a function of s. $\square$

Definition 3.2 (Pressure-normalized critical quantities)**.** For $z_0 = (x_0, t_0)$ and $r > 0$, set

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3\,dx\,dt$$

and

$$D(z_0, r) = r^{-2} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |p(x,t) - (p)_{B_r(x_0)}(t)|^{3/2}\,dx\,dt.$$

Here $(p)_{B_r(x_0)}(t)$ denotes the spatial mean of $p(\cdot, t)$ on $B_r(x_0)$.

The subtraction of the spatial mean fixes the pressure ambiguity in the form used by the Caffarelli–Kohn–Nirenberg criterion.

Proposition 3.3 (Local finiteness and scaling). *Let (u,p) be a suitable weak solution near $z_0 = (x_0, T)$. Then $C(z_0, r)$ and $D(z_0, r)$ are finite for every sufficiently small r. Moreover, for every fixed $R < \infty$,*

$$\int_{-R^2}^{0} \int_{B_R} \Big(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \Big)\, dy\, ds$$
$$= R^2 \big(C(z_0, rR) + D(z_0, rR) \big).$$

Proof. Local finiteness of the velocity term follows from

$$u \in L^\infty_t L^2_{x,\mathrm{loc}} \cap L^2_t H^1_{x,\mathrm{loc}} \subset L^3_{\mathrm{loc}}$$

on finite cylinders, using the local Sobolev interpolation. The pressure term is finite because $p \in L^{3/2}_{\mathrm{loc}}$, and subtracting a spatial mean over the same ball preserves local $L^{3/2}$ integrability.

For the scaling identity, use $x = x_0 + ry$, $t = T + r^2 s$. The velocity term transforms as

$$\int_{-R^2}^{0} \int_{B_R} |u^{(r)}|^3 \, dy\, ds = r^{-2} \int_{T-(rR)^2}^{T} \int_{B_{rR}(x_0)} |u|^3 \, dx\, dt.$$

The pressure means transform according to

$$(p^{(r)})_{B_R}(s) = r^2 (p)_{B_{rR}(x_0)}(T + r^2 s),$$

and the pressure term gives the corresponding identity for D. □

4. Epsilon Regularity and the Vanishing Case

Theorem 4.1 (Caffarelli–Kohn–Nirenberg epsilon regularity). *There exists a universal constant $\varepsilon_{\mathrm{CKN}} > 0$ with the following property. Let (u,p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) + D(z_0, r) < \varepsilon_{\mathrm{CKN}},$$

then u is locally bounded in a smaller cylinder, for instance in $Q_{r/2}(z_0)$. Consequently z_0 is a regular point.

Proof. This is the local epsilon-regularity theorem of Caffarelli–Kohn–Nirenberg [1], stated with the pressure normalization used in D. Equivalent representatives of the pressure differ by functions of time, and the spatial mean subtraction on $B_r(x_0)$ removes this ambiguity. □

Definition 4.2 (Vanishing local critical density at time T). We say that the local critical density vanishes at time T if, for every $x_0 \in \Sigma(T)$,

$$\limsup_{r \downarrow 0} \big(C((x_0, T), r) + D((x_0, T), r) \big) = 0.$$

Proposition 4.3 (Equivalent rescaled formulation). *The preceding condition is equivalent to the following statement: for every $x_0 \in \Sigma(T)$ and every fixed $R < \infty$,*

$$\lim_{r \downarrow 0} \int_{-R^2}^{0} \int_{B_R} \Big(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \Big)\, dy\, ds = 0.$$

Proof. By Proposition 3.3, the rescaled integral over $B_R \times (-R^2, 0)$ equals

$$R^2 \big(C((x_0, T), rR) + D((x_0, T), rR) \big).$$

Thus vanishing of $C + D$ implies vanishing on every fixed rescaled cylinder. Conversely, taking $R = 1$ gives the original pointwise condition. □

Theorem 4.4 (No-concentration regularity). *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. If the local critical density vanishes at time T, then $\Sigma(T) = \emptyset$. Thus no finite-time singularity occurs at time T in the no-concentration case.*

Proof. Suppose, toward a contradiction, that $x_0 \in \Sigma(T)$. By the vanishing hypothesis, choose $r > 0$ such that
$$C((x_0,T),r) + D((x_0,T),r) < \varepsilon_{\mathrm{CKN}}.$$
Theorem 4.1 implies that (x_0,T) is regular. This contradicts the definition of $\Sigma(T)$. Hence $\Sigma(T) = \emptyset$. $\square$

Corollary 4.5 (Vanishing rescaled density implies regularity)**.** *Let (u,p) be suitable near (x_0,T). If*
$$\limsup_{r\downarrow 0}\big(C((x_0,T),r) + D((x_0,T),r)\big) = 0,$$
then (x_0,T) is regular. Equivalently, a singularity at (x_0,T) requires nonzero pressure-normalized CKN density on a sequence of shrinking backward parabolic cylinders.

Proof. The hypothesis gives a scale at which $C+D$ is below the universal threshold in Theorem 4.1. The epsilon-regularity theorem gives local boundedness near (x_0,T). $\square$

5. Positive Local Concentration

Lemma 5.1 (Failure of vanishing gives a positive concentration sequence)**.** *Suppose the local critical density does not vanish at time T. Then there exist $x_0 \in \Sigma(T)$, a number $\eta > 0$, and radii $r_n \downarrow 0$ such that*
$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \textit{for all } n.$$
Equivalently, the rescaled sequence around (x_0,T) has nonzero critical local density on a fixed bounded cylinder.

Proof. Negating the definition of vanishing gives a point $x_0 \in \Sigma(T)$ with positive limsup of
$$C((x_0,T),r) + D((x_0,T),r)$$
as $r \downarrow 0$. Choose $\eta > 0$ below that limsup and choose a sequence $r_n \downarrow 0$ along which the displayed lower bound holds. The rescaled formulation follows from Proposition 3.3. $\square$

Definition 5.2 (Positive local concentration sequence)**.** Let $x_0 \in \Sigma(T)$. A sequence $r_n \downarrow 0$ is a positive local concentration sequence at (x_0,T) if there is $\eta > 0$ such that
$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \text{for all } n.$$
The corresponding parabolic rescalings are those defined by
$$u_n(y,s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Proposition 5.3 (Persistence of positive concentration)**.** *Let (u_n,p_n) be suitable weak solutions on a fixed cylinder $B_R \times (-R^2,0)$. Suppose that*
$$\int_{-R^2}^{0}\int_{B_R}\Big(|u_n|^3 + |p_n - (p_n)_{B_R}(s)|^{3/2}\Big)\,dy\,ds \geq \eta > 0$$
for all n. Suppose further that, after passing to a subsequence,
$$u_n \to U \quad \textit{in } L^3(B_R \times (-R^2,0))$$
and
$$p_n - (p_n)_{B_R}(s) \to \Pi \quad \textit{in } L^{3/2}(B_R \times (-R^2,0)).$$
Then
$$\int_{-R^2}^{0}\int_{B_R}\Big(|U|^3 + |\Pi|^{3/2}\Big)\,dy\,ds \geq \eta.$$
In particular, the limiting pair has nonzero pressure-normalized local critical density on this cylinder. If $\Pi = P - (P)_{B_R}(s)$, this is the CKN density of (U,P) on $B_R \times (-R^2,0)$.

Proof. Strong convergence in L^3 gives

$$\int_{-R^2}^{0}\int_{B_R}|u_n|^3\,dy\,ds \longrightarrow \int_{-R^2}^{0}\int_{B_R}|U|^3\,dy\,ds.$$

Strong convergence in $L^{3/2}$ gives the analogous convergence of the normalized pressure integrals. Passing to the limit in the lower bound yields the asserted inequality. ☐

Proposition 5.4 (Local alternative)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$ for some $\delta>0$. Exactly one of the following alternatives holds.*

(i) *For every $x_0\in\Sigma(T)$,*

$$\limsup_{r\downarrow 0}\big(C((x_0,T),r)+D((x_0,T),r)\big)=0.$$

(ii) *There exist $x_0\in\Sigma(T)$, $\eta>0$, and $r_n\downarrow 0$ such that*

$$C((x_0,T),r_n)+D((x_0,T),r_n)\geq\eta \qquad \textit{for all } n.$$

Proof. The first alternative is a universal assertion over the singular set. If it fails, then for some $x_0\in\Sigma(T)$ the corresponding limsup is positive; choosing a positive number below this limsup and a realizing sequence gives the second alternative. Conversely, the second alternative contradicts the first. ☐

Theorem 5.5 (Singular points carry positive local CKN concentration)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$. If $\Sigma(T)\neq\emptyset$, then there are $x_0\in\Sigma(T)$, $\eta>0$, and $r_n\downarrow 0$ such that*

$$C((x_0,T),r_n)+D((x_0,T),r_n)\geq\eta \qquad \textit{for all } n.$$

Proof. If the first alternative in Proposition 5.4 held, then Theorem 4.4 would imply $\Sigma(T)=\emptyset$, contrary to the hypothesis. Therefore the second alternative in Proposition 5.4 holds. ☐

6. The Type I/Type II Blowup-Rate Dichotomy

Let $x_0\in\Sigma(T)$, $\eta>0$, and $r_n\downarrow 0$ be given by Theorem 5.5. Define

$$u_n(y,s)=r_n u(x_0+r_n y, T+r_n^2 s), \qquad p_n(y,s)=r_n^2 p(x_0+r_n y, T+r_n^2 s).$$

Then (u_n,p_n) are suitable weak solutions on expanding backward cylinders, and by Proposition 3.3 they retain a nonzero pressure-normalized critical density on a fixed bounded cylinder.

Definition 6.1 (Type I concentration sequence)**.** Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$, and let $r_n\downarrow 0$ be a positive local concentration sequence at (x_0,T). The sequence r_n is called Type I if it satisfies a local Type I scale-invariant bound at (x_0,T): there are $\rho>0$ and $M<\infty$, with $\rho^2<\delta$, such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T}\sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_0))}\leq M.$$

The corresponding case is the Type I alternative.

Definition 6.2 (Local Type II alternative)**.** Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$, and let $r_n\downarrow 0$ be a positive local concentration sequence at (x_0,T). The sequence r_n is in the local Type II alternative if no local Type I scale-invariant bound holds at (x_0,T). Equivalently, for every $\rho>0$ with $\rho^2<\delta$,

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T}\sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_0))}=\infty.$$

Theorem 6.3 (Local blowup-rate dichotomy)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then there exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \text{for all } n.$$

The sequence r_n is a positive local concentration sequence. It is either a Type I concentration sequence or a local Type II alternative. Consequently, after the no-concentration case has been excluded by epsilon regularity, every finite-time singular case reaches exactly one of the two blowup-rate alternatives.

Proof. If the local critical density vanished at every point of $\Sigma(T)$, then Theorem 4.4 would give $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore Theorem 5.5 gives $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ with the stated lower bound. The rescaled sequence has positive local critical density by Proposition 3.3. It either satisfies the Type I bound in Definition 6.1, or it does not. In the latter case it is, by Definition 6.2, a local Type II alternative. ☐

Remark 6.4. The preceding theorem is local. It does not assert global regularity, scattering, profile classification, Type I rigidity, or exclusion of Type II concentration. It proves only the local CKN concentration step and the ensuing blowup-rate dichotomy.

References

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